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import math

# Part A – Machine Initialization
machines = ['A', 'B', 'C']
N_play = 10

# Part B – Initialize N, R, Q and UCB
N = {'A': 0, 'B': 0, 'C': 0}
R = {'A': 0, 'B': 0, 'C': 0}
Q = {'A': 0, 'B': 0, 'C': 0}
UCB = {'A': 0, 'B': 0, 'C': 0}

# Part C – UCB Calculation
def calculate_ucb(machine, play):

    if N[machine] == 0:
        return float('inf')

    return Q[machine] + math.sqrt(
        (2 * math.log(play)) / N[machine]
    )

def select_machine(play):

    for machine in machines:
        UCB[machine] = calculate_ucb(machine, play)

    return max(UCB, key=UCB.get)

# Part D, E – Reward Collection and Q Update
for play in range(1, N_play + 1):

    selected_machine = select_machine(play)

    print("\nPlay", play)
    print("UCB values:", UCB)
    print("Selected Machine:", selected_machine)

    reward = float(input("Enter reward: "))

    # Update N
    N[selected_machine] += 1
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# Update total reward
R[selected_machine] += reward

# Incremental Q-value update
Q[selected_machine] = (
    Q[selected_machine]
    + (reward - Q[selected_machine]) / N[selected_machine]
)

# Part F - Display Results
print("\n===== FINAL RESULTS =====")

print("Machine\tN\tR\tQ-value\tUCB")

for machine in machines:

    UCB[machine] = calculate_ucb(machine, N_play)

    print(
        f"{machine}\t"
        f"{N[machine]}\t"
        f"{R[machine]:.2f}\t"
        f"{Q[machine]:.2f}\t"
        f"{UCB[machine]:.2f}"
    )

best_machine = max(Q, key=Q.get)

print("\nBest Machine:", best_machine)
print("Highest Q-value:", round(Q[best_machine], 2))
print("Total Reward:", sum(R.values()))55

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Play 1
UCB values: {'A': inf, 'B': inf, 'C': inf}
Selected Machine: A
Enter reward: 8

Play 2
UCB values: {'A': 9.177410022515474, 'B': inf, 'C': inf}
Selected Machine: B
Enter reward: 1

Play 3
UCB values: {'A': 9.482303807367511, 'B': 2.4823038073675114, 'C':
Selected Machine: C
Enter reward: 5

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Play 4
UCB values: {'A': 9.665109222315396, 'B': 2.6651092223153956, 'C': 6.6651092223153956}
Selected Machine: A
Enter reward: 5

Play 5
UCB values: {'A': 7.76863624117952, 'B': 2.7941225779941012, 'C': 6.7941225779941012}
Selected Machine: A
Enter reward: 5

Play 6
UCB values: {'A': 7.092934724866359, 'B': 2.8930184728248456, 'C': 6.8930184728248456}
Selected Machine: A
Enter reward: 5

Play 7
UCB values: {'A': 6.736384851124376, 'B': 2.972769702248751, 'C': 6.972769702248751}
Selected Machine: C
Enter reward: 5

Play 8
UCB values: {'A': 6.769666990168809, 'B': 3.039333980337618, 'C': 6.039333980337618}
Selected Machine: A
Enter reward: 5

Play 9
UCB values: {'A': 6.537491243124162, 'B': 3.09629414793641, 'C': 6.09629414793641}
Selected Machine: A
Enter reward: 5

Play 10
UCB values: {'A': 6.3760869616261555, 'B': 3.145966026289347, 'C': 6.145966026289347}
Selected Machine: C
Enter reward:

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ValueError                                Traceback (most recent
call last)
/tmp/ipykernel_1809/1054242515.py in <cell line: 0>()
    40     print("Selected Machine:", selected_machine)
    41
--> 42     reward = float(input("Enter reward: "))
    43
    44     # Update N

```

Next steps: [Explain error](#)

